Vietnamese PII NER
Error Analysis Report

1. Executive Summary

- Validation set: 6,014 samples, including 3,360 samples with at least one PII entity and 2,654 samples without entity.
- Overall ranking: PhoBERT ≈ XLM-RoBERTa > BiLSTM-CRF > BiLSTM.
- XLM-RoBERTa has the best aggregate entity-level F1 in the README table, while PhoBERT and XLM-RoBERTa are almost tied in per-sample analysis.
- The main remaining error type is wrong_type: the model often detects that a token is PII, but assigns the wrong PII subtype.
- BiLSTM-CRF improves over BiLSTM because CRF adds sequence-level constraints for BIO tagging.

2. Model-Level Summary

Model	Mean sample F1	Exact all	Mean F1 entity samples	Exact entity samples	Exact no-entity samples
BiLSTM	0.9589	88.58%	0.9283	79.73%	99.77%
BiLSTM-CRF	0.9685	91.02%	0.9451	84.08%	99.81%
PhoBERT	0.9782	93.50%	0.9610	88.36%	100.00%
XLM-RoBERTa	0.9780	93.35%	0.9610	88.12%	99.96%

Table 1. Per-sample F1 and exact-match summary.

Interpretation. Exact-match is stricter than F1: a sample is exact-match only when the entire BIO sequence is predicted correctly. Therefore, exact-match is useful for finding hard samples, but it should be interpreted together with F1.

3. What Types of Errors Remain?

The token-level analysis groups model mistakes into four practical categories:

- missed_entity: the true label is an entity, but the model predicts O.
- false_positive: the true label is O, but the model predicts an entity.
- wrong_type: both true and predicted labels are entities, but the entity type is different.
- boundary_error: the type is correct, but the BIO prefix or span boundary is inconsistent.

Error type	BiLSTM	BiLSTM-CRF	PhoBERT	XLM-RoBERTa
missed_entity	218	157	92	43
false_positive	291	263	174	141
wrong_type	1516	1296	906	1027
boundary_error	43	33	21	21

Table 2. Token-level error counts by model.

Main takeaway. wrong_type is the dominant error group. This means the models usually recognize that a token belongs to PII, but struggle to distinguish closely related PII subtypes.

4. Main Confusion Patterns

The most important wrong-type patterns are caused by PII types with similar surface forms:

- DATE vs DOB: both look like date strings; the difference often depends on context such as “sinh ngày” or “hết hạn vào ngày”.
- MASKEDNUMBER vs CREDITCARDNUMBER: both can appear as long numeric sequences.
- LITECOINADDRESS vs BITCOINADDRESS: both are long crypto-address-like strings with limited linguistic context.
- JOBTYPE vs JOBTITLE: both describe work-related information and can be semantically close.
- FIRSTNAME vs MIDDLENAME: name components are difficult when the boundary or Vietnamese name order is ambiguous.

5. Difficult Entity Types

Entity	Support	BiLSTM	BiLSTM-CRF	PhoBERT	XLM-R	Avg F1
DOB	178	0.6006	0.7152	0.7262	0.6271	0.6673
DATE	334	0.7787	0.8526	0.8367	0.8264	0.8236
JOBTYPE	196	0.8105	0.8385	0.8663	0.8564	0.8429
CURRENCYCODE	65	0.8217	0.8372	0.8824	0.8507	0.8480
MASKEDNUMBER	178	0.7607	0.8608	0.8954	0.8802	0.8493
CREDITCARDNUMBER	183	0.7680	0.8580	0.9003	0.8900	0.8541
MIDDLENAME	209	0.8550	0.8536	0.8889	0.8868	0.8711
JOBTITLE	241	0.8481	0.8821	0.8942	0.8902	0.8787
COMPANYNAME	193	0.8384	0.8714	0.9460	0.9223	0.8945
SEX	161	0.9048	0.9102	0.9268	0.9518	0.9234

Table 3. Lowest average F1 entity types across four models.

Interpretation. The hardest groups are not random. They are mostly entities that require context to distinguish, or entities whose surface forms overlap with other PII categories.

6. PhoBERT vs XLM-RoBERTa

PhoBERT and XLM-RoBERTa are very close overall. The better conclusion is not “PhoBERT is worse” or “XLM-R dominates”, but that Vietnamese PII NER depends on two kinds of signals:

- Vietnamese linguistic context, where PhoBERT can be competitive because it is specialized for Vietnamese.
- Universal PII patterns such as emails, IP addresses, usernames, URLs, dates, account-like numbers, and crypto addresses, where XLM-RoBERTa can remain very strong despite being multilingual.

Entity	Support	PhoBERT	XLM-R	Diff
DOB	178	0.7262	0.6271	0.0992
CURRENCYCODE	65	0.8824	0.8507	0.0316
COMPANYNAME	193	0.9460	0.9223	0.0237
MASKEDNUMBER	178	0.8954	0.8802	0.0152
LITECOINADDRESS	70	0.9683	0.9552	0.0130
DATE	334	0.8367	0.8264	0.0103
CREDITCARDNUMBER	183	0.9003	0.8900	0.0103
JOBTYPE	196	0.8663	0.8564	0.0100

Table 4. Entity types where PhoBERT is ahead of XLM-RoBERTa.

Entity	Support	PhoBERT	XLM-R	Diff
HEIGHT	54	0.9455	0.9725	0.0270
SEX	161	0.9268	0.9518	0.0250
USERNAME	234	0.9787	0.9979	0.0191
AGE	174	0.9623	0.9797	0.0174
SECONDARYADDRESS	163	0.9695	0.9848	0.0153
CITY	204	0.9853	1.0000	0.0147
CREDITCARDISSUER	109	0.9722	0.9863	0.0141
MAC	88	0.9884	1.0000	0.0116

Table 5. Entity types where XLM-RoBERTa is ahead of PhoBERT.

7. Why BiLSTM-CRF Improves BiLSTM

BiLSTM predicts token labels from sequential representations. CRF adds a sequence-level decoding layer that learns transitions between labels. This matters because BIO labels are not independent.

- A transition such as B-DATE -> I-DATE is reasonable.
- A transition such as O -> I-DATE is usually invalid or suspicious.
- Therefore, CRF can reduce inconsistent spans and some boundary-related mistakes.

Error type	BiLSTM	BiLSTM-CRF	Reduction
missed_entity	218	157	61
false_positive	291	263	28
wrong_type	1516	1296	220
boundary_error	43	33	10

Table 6. Error reduction from BiLSTM to BiLSTM-CRF.

8. Representative Case Studies

all models correct

Sample 1: Tôi đã nhận thấy một sự không nhất quán trong các báo cáo lưu lượng truy cập cho các chiến dịch tiếp thị kỹ thuật số của chúng ta. Liên, điều này có vẻ liên quan đến địa ...

F1: BiLSTM=1.00, BiLSTM-CRF=1.00, PhoBERT=1.00, XLM-R=1.00

Explanation: Entity có pattern/ngữ cảnh rõ nên cả recurrent và transformer đều nhận diện đúng.

all models wrong

Sample 6: Xin chào Tuyết Thị, bạn đã đăng ký được cho chuẩn bị cho giai đoạn huấn luyện chuyên nghiệp huấn số. Chúng tôi đã ghi nhận 70 535377 tiền vào tài khoản của bạn để chi trả...

F1: BiLSTM=0.67, BiLSTM-CRF=0.67, PhoBERT=0.86, XLM-R=0.67

Explanation: Mẫu có ranh giới hoặc subtype dễ nhầm; các model đều không exact-match.

bilstm crf fixes bilstm

Sample 10: Ráng buộc mình vui mừng được bạn, Thị. Xin vui lòng thanh toán Bảng Anh số 7826146759451784 đến tài khoản của chúng tôi để hoàn tất phiếu điều trị thuốc tính của bạn. Xác...

F1: BiLSTM=0.80, BiLSTM-CRF=1.00, PhoBERT=1.00, XLM-R=1.00

Explanation: CRF giúp chuỗi BIO nhất quán hơn nên sửa được lỗi của BiLSTM.

transformers better than recurrent

Sample 83: Chào, cho buổi thuyết trình sắp tới, chúng tôi cần một báo cáo chi tiết minh họa hiệu suất của VND so với các loại tiền tệ tương tự trên thế giới. Vui lòng sử dụng các nguồn dữ liệu chính xác và theo dõi email sang.mai729@fpt.vn.

F1: BiLSTM=0.50, BiLSTM-CRF=0.50, PhoBERT=1.00, XLM-R=1.00

Explanation: Transformer tận dụng pretrained context tốt hơn trong câu dài/nhiều entity.

phobert better than xlmr

Sample 6: Xin chào Tuyết Thị, bạn đã đăng ký được cho chuẩn bị cho giai đoạn huấn luyện chuyên nghiệp huấn số. Chúng tôi đã ghi nhận 70 535377 tiền vào tài khoản của bạn để chi trả...

F1: BiLSTM=0.67, BiLSTM-CRF=0.67, PhoBERT=0.86, XLM-R=0.67

Explanation: PhoBERT nhỉnh hơn ở mẫu cần ngữ cảnh tiếng Việt hoặc cách diễn đạt tự nhiên.

xlmr better than phobert

Sample 27: Khuê, theo hồ sơ của chúng tôi, thành viên của bạn cho Hiệp hội Luật Nhân quyền sẽ hết hạn vào ngày 17-10-1992. Chi phí gia hạn là VND594658.70. Chúng tôi cung cấp nhiều ...

F1: BiLSTM=0.83, BiLSTM-CRF=1.00, PhoBERT=0.83, XLM-R=1.00

Explanation: XLM-R nhỉnh hơn ở mẫu nhiều pattern phổ quát như mã, số, ngày tháng, crypto/email.

9. Conclusion

Recommended conclusion: Most remaining errors do not come from the model completely missing PII. They mainly come from confusion between visually similar PII types and from boundary/span mistakes. Transformer-based models achieve the best results because of stronger contextual representations, while BiLSTM-CRF improves over BiLSTM by enforcing more consistent BIO label transitions. PhoBERT and XLM-RoBERTa are almost equivalent on this task, suggesting that Vietnamese PII NER relies on both Vietnamese context and universal PII patterns.